

where similar developments will follow. Moreover, radiology is a highly-paid medical specialty, which means that the potential benefits from productivity enhancements through AI are large. Radiology is also ideal from a research perspective. Unlike other physicians, radiologists do not have direct contact with patients, allowing us to run a decision experiment that resembles their normal workflow through a remote interface that we developed. To aid quantitative analysis, instead of obtaining a free-text report, we collect radiologist’s assessed probability that a given chest pathology is present and a binary clinical recommendation of whether to treat or follow up for that pathology. For our experiment, we hired 180 radiologists through teleradiology companies that serve US hospitals and offer the services of both US-based and non-US-based radiologists.

Analyzing the quality of our participants’ assessments requires establishing, for each patient case, a standard or ground truth. An important challenge in our setting is that definitive diagnostic tests do not exist for most thoracic pathologies and, even when they do exist, are selectively performed depending on a radiologist’s recommendation. We therefore follow the machine learning literature (Sheng et al., 2008) and construct a diagnostic standard by aggregating the assessments of five board-certified radiologists at Mt. Sinai Hospital with at least ten years of experience and chest radiology as a sub-specialty. We assess the robustness of all our results by constructing a (leave-one-out) ground truth using the assessments of our experimental participants and by varying the aggregation method.

There are two important challenges in designing the experiment, which we address using a combination of experimental designs. First, while teleradiology firms allow us to recruit radiologists for our experiment we need to compensate them at the market rate, making data collection expensive. An across-participant design would therefore be cost-prohibitive except for extremely large effect sizes. In the first design, we, therefore, randomize each participant into an order in which they are exposed to the four informational environments but do not encounter cases repeatedly. Thus, this design allows for within-participant comparisons in addition to an initial pure across-participant comparison. The second challenge is that, to estimate a model of belief updating, we need to observe radiologists’ assessments of a given case both with and without AI assistance. Moreover, the assessment without AI assistance is required to estimate the Bayesian benchmark, which we can compare to the assessment that incorporates AI assistance. Our second design addresses this issue by asking each participant to diagnose each patient case once in each of the four information environments in random order while ensuring at least a two-week wash-out period between two encounters of a given case. This wash-out period is intended to eliminate the memory of prior information and

$\omega_i = 1$ to $\omega_i = 0$ given that the human’s signal is s_{ir}^E . The first term is positive if, given the realization s_{ir}^E , the signal s_i^A is more likely if $\omega_i = 1$ as compared to $\omega_i = 0$. In this case, the posterior odds shift in favor of the state $\omega_i = 1$. Analogously, the odds shift away from $\omega_i = 1$ if, given s_{ir}^E , the signal s_i^A is more likely if the state were $\omega_i = 0$.

The task of empirically analyzing deviations in participants’ beliefs from the benchmark in equation (3) is challenging because the signals differ across cases i and humans may, due to differences in skill, have heterogeneous signal distributions $\pi_r(\cdot)$. The ideal dataset would elicit $p_r(\omega_i = 1 \mid s_i^A, s_{ir}^E)$ and $\pi_r(\omega_i = 1 \mid s_{ir}^E)$ for the same case and the same human to keep the signals (s_i^A, s_{ir}^E) , and human-specific parameters fixed, and use the latter to estimate $\pi_r(\omega_i = 1 \mid s_i^A, s_{ir}^E)$ using equation (3).

However, empirically implementing this strategy requires us to confront several experimental and methodological challenges. First, the experiment needs to be sensitive to anchoring and order effects when eliciting $p_r(\omega_i = 1 \mid s_i^A, s_{ir}^E)$ and $\pi_r(\omega_i = 1 \mid s_{ir}^E)$ from participants. Second, the potential for measurement error and the dependence of the first term in equation (3) on a latent signal s_{ir}^E complicates the exercise. We defer our approach to this second challenge to section 5, and turn our attention to the experimental design.

3 Setting and Experiment

Our experiment elicits the probability of a pathology’s presence $p_r(\omega = 1 \mid s)$ and a recommended treatment/follow-up decision a under varying information treatments. There are four information treatments in the experiment, with only the chest X-ray in the minimal information case, to which we add AI assistance, contextual information, or both. Next, we describe the experimental context and interface before presenting the design of our experiments.

3.1 Experimental Context

3.1.1 Radiology

Radiologists diagnose the presence of a given pathology at the request of a treating physician. The information available to a radiologist consists of diagnostic images (e.g. chest X-rays), any relevant medical history (e.g. laboratory results), and clinical indication notes of the treating physician.⁸ The treating physician’s notes are of varying detail levels – they may provide no clinical information or guidance, request the analysis of a specific pathology, or

⁸Radiologists are rarely in direct contact with the patient or the treating physician, except via the formal information exchange outlined here.

only list the patient’s primary symptom (see appendix B.2 for examples). Irrespective of the pathology suspected by the treating physicians, radiologists are expected to report all pathological findings.

Because image-based classification is a core task performed by radiologists, a high-paying profession, it is not surprising that AI tools have made significant inroads in the field in the last decade. Recent advances in deep learning methods for image recognition have yielded algorithms that can match or surpass the performance of human radiologists (Obermeyer and Emanuel, 2016; Langlotz, 2019). As of 2020, 55 companies offered a total of 119 algorithmic products of which 46 have FDA approval (Tadavarthi et al., 2020). Most products related to clinical decision-making are marketed as support tools as opposed to autonomous tools, partly due to regulatory and liability issues (Harvey and Gowda, 2020).

3.1.2 *CheXpert*

We provide AI assistance using the CheXpert model, which is a deep learning prediction algorithm for chest X-ray interpretation (Irvin et al., 2019). This model is trained on a dataset of 224,316 chest radiographs of 65,240 patients labeled for the presence of fourteen common chest radiographic pathologies.⁹ The algorithm does not use any other patient information, such as the clinical history or vitals.¹⁰ Nonetheless, a prior version of this algorithm was shown to match or surpass the performance of board-certified radiologists from Stanford Hospital on five pathologies (Patel et al., 2019). These study results are also presented to our participants when introducing the AI tool. Section 4 confirms that the algorithm outperforms a majority of radiologists in our experiment. We relegate additional details about the algorithm to appendix B.3. The algorithm assistance to our participants will be in the form of a vector of probabilities for the presence of every pathology.¹¹

⁹The term artificial intelligence is typically reserved for a system of different prediction tasks to mimic a more complex set of behaviors, whereas machine learning is concerned with one specific prediction task. For a detailed discussion of this distinction see, for instance, (Taddy, 2018).

¹⁰While large datasets of images are increasingly available (e.g. Kramer et al. 2011, Johnson et al. 2016, Irvin et al. 2019) it is significantly more difficult to construct such datasets for other patient information due to the compulsory manual review of textual data for HIPPA compliance.

¹¹Some algorithms attempt to make their predictions explainable to a human by highlighting the parts of the image that drive a specific prediction. However, prior studies show that providing such localization in addition to the numeric output does not improve the accuracy of radiologists (Gaube et al., 2022). A quantitative output allows us to compute a Bayesian benchmark to the radiologist’s prediction, which is otherwise difficult.

highly-skilled profession that is being transformed by AI, radiology also represents a large class of professionals whose main job is a high-stakes classification task. Since we can simulate radiologists’ normal workflow, this is an ideal setting for conducting such an experiment.

While the potential benefits of deploying AI assistance are large in our setting, biases in humans’ use of AI assistance eliminate these potential gains. Even though the AI tool in our experiment performs better than two thirds of radiologists, optimally combining their predictions could substantially improve performance. Yet, we find that giving radiologists access to AI predictions does not, on average, lead to higher performance. This average treatment effect, however, masks systematic heterogeneity: providing AI does improve radiologists’ predictions and decisions for cases where the AI is certain, but not when uncertain. This latter result – that prediction quality can be reduced for some range of AI signals – rejects Bayesian updating with correct beliefs. We also describe systematic errors in belief updating – namely radiologists exhibit automation neglect (e.g. radiologists weight the AI prediction relative to their own) and treat the AI prediction and their own signals as independent even though they are not. Moreover, radiologists take significantly more time to make a decision when AI information is provided.

Together, these results have important implications for how to design the collaboration between humans and machines. Increased time costs and sub-optimal use of AI information both work against having radiologists make decisions with AI assistance. In fact, an optimal delegation policy that utilizes heterogeneity in treatment effects given the AI prediction suggests that cases should either be decided by the AI alone or by the radiologist alone. Only a small share of cases are optimally delegated to radiologists with access to AI. In other words, we find that radiologists should work *next to* as opposed to *with* AI. To the extent that expert decision makers generally under-respond to information other than their own (Conlon et al., 2022) and incorporating additional information is cognitively costly, these insights may hold in other settings where experts’ main job is a classification task.

There are several important considerations that are outside the scope of this work. One question motivated by the unrealized potential gains of AI assistance we find is whether the benefits from AI-specific training for radiologists and/or experience with AI are large. Answering these questions requires different experimental designs or longer-run studies. Another open question is whether the heterogeneity in the value of AI assistance is correlated with a human’s baseline skill or other characteristics, and whether such correlation can be predicted to target assistance. A consideration in the organization of human-AI collaboration that we abstract away from is that the form of collaboration may influence experts’ incentives, to which humans may respond strategically. Finally, the use of AI in practice will be